

Error Log

Noorinder

1. Missing Override Annotation

- **Error:** The `getDetails()` method in subclasses wasn't marked with `@Override`, making it unclear that the method was overriding an abstract method.
- **Cause:** The annotation was omitted during implementation.
- **Fix:** Added the `@Override` annotation to explicitly indicate the method's inheritance relationship.

Before Fix:

java

Copy code

```
public String getDetails() {  
    return String.format("Car: %s, Doors: %d", super.toString(), numDoors);  
}
```

After Fix:

java

Copy code

```
@Override  
public String getDetails() {  
    return String.format("Car: %s, Doors: %d", super.toString(), numDoors);  
}
```

2. Handling Invalid Vehicle Type

- **Error:** Entering an invalid vehicle type caused incorrect behavior or infinite loops.
- **Cause:** The `switch` statement didn't handle invalid inputs.
- **Fix:** Added a `default` case to display an error message and re-prompt the user.

Before Fix:

```

switch (vehicleType) {
case 1:
    vehicle = car;
    break;
case 2:
    vehicle = truck;
    break;
case 3:
    vehicle = minivan;
    break;
// No handling for invalid input
}

```

After Fix:

```

switch (vehicleType) {
case 1:
    vehicle = car;
    break;
case 2:
    vehicle = truck;
    break;
case 3:
    vehicle = minivan;
    break;
default:
    System.out.println("Invalid Vehicle type! Please try again.");
    continue; // Re-prompts the user
}

```

3. Lack of Validation for Vehicle Year

- **Error:** The program allowed invalid year values, such as negative years or years beyond the current date.
- **Cause:** The `Vehicle` constructor didn't validate the year input.
- **Fix:** Added validation in the `Vehicle` constructor to ensure the year is realistic.

Before Fix:

```

public Vehicle(String make, String model, int year) {
    this.make = make;
    this.model = model;
    this.year = year;
}

```

After Fix:

```
public Vehicle(String make, String model, int year) {  
    if (year < 1886 || year > 2024) { // Validate year  
        throw new IllegalArgumentException("Invalid year for a vehicle.");  
    }  
    this.make = make;  
    this.model = model;  
    this.year = year;  
}
```

4. Incomplete Vehicle Details

- **Error:** The `getDetails()` method in `Minivan` didn't include the seating capacity, leading to incomplete output.
- **Cause:** The `getDetails()` method was missing the seating capacity field.
- **Fix:** Updated the method to include seating capacity.

Before Fix:

```
@Override  
public String getDetails() {  
    return super.toString(); // Missing seating capacity  
}
```

After Fix:

```
@Override  
public String getDetails() {  
    return String.format("Minivan: %s, Seating Capacity: %d", super.toString(), seatingCapacity);  
}
```

5. No Quit Confirmation

- **Error:** The program exited immediately when the quit option was selected, without any confirmation.
- **Cause:** There was no confirmation prompt for quitting.
- **Fix:** Added a confirmation prompt before exiting.

Before Fix:

```
if (action.equalsIgnoreCase("Q")) {  
    break; // Immediate exit  
}
```

After Fix:

```
if (action.equalsIgnoreCase("Q")) {  
    System.out.print("Are you sure you want to quit? (yes/no): ");  
    String confirmation = input.next();  
    if (confirmation.equalsIgnoreCase("yes")) {  
        break; // Exit the loop  
    }  
}
```